# Amos Roche

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## EDUCATION

**Columbia University** Sep 2023 - Dec 2024
*Master of Science in Data Science* *New York City, NY*

**Worcester Polytechnic Institute** Aug 2020 - May 2023
*Bachelor of Science in Data Science* *Worcester, MA*

## WORK EXPERIENCE

**Kaliber Labs** Mar 2025 - Present
*AI Engineer* *San Francisco, CA*

- Led end-to-end development of a **TTS System**, assembling **10M+ speech clips** and implementing **conditional flow-matching** and **state-space** architectures to improve expressiveness and naturalness.
- Developed a **hierarchical multi-task learning** architecture using **WavLM** and **cross-attention**, achieving state-of-the-art (SOTA) performance on L2-ARCTIC with **13.47% phoneme error rate** through articulatory feature supervision; manuscript in preparation.
- Built and productionized an **AI document-intake pipeline** for hospital systems serving **300+ patients** and **50+ providers**.
- Designed **context-engineering pipelines for a 4-bit quantized LLM on edge hardware**, structuring conversation history and patient-state data to drive digital-twin reasoning and updates.

**Walmart Connect** Jun 2024 - Aug 2024
*Data Science Intern (Ads)* *New York City, NY*

- Identified **control/test-group mismatches in ~13% of ad campaigns** using A/A testing and correlation analysis, driving updates to campaign shareability criteria and preventing inaccurate sales-lift reporting to advertisers.
- Built **PySpark/SQL data pipelines** to automate daily time-series ingestion from Hive to GCP, improving data-pull performance by 7.8%

**Fidelity Investments** Jun 2023 - Aug 2023
*Data Engineering Intern* *Durham, NC*

- Built Python/Groovy pipelines from **AWS S3** to **Snowflake** and migrated transformations from **ETL to dbt-based ELT,** improving efficiency by 6.7%; automated access and 30-day retention policies with **CloudFormation**

**Fidelity Investments (Fidelity Center for Applied Technology)** Jun 2022 - Aug 2022
*Data Science Intern* *Boston, MA*

- Developed an **NLP email-classification pipeline for 60+ business units**, improving performance for **55/60 units** using Linear SVC and advanced sampling techniques; built a **Flask front end** for comparing embeddings, sampling strategies, and model configurations

**National Science Foundation Data Science REU** May 2021 - Jan 2022
*Machine Learning Researcher* *Worcester, MA*

- Developed a **dynamic model-stacking framework** using convex optimization that outperformed baselines on **6/9 behavioral outcomes** for childhood cancer survivors; built a neural-network-based feature selection.

## RESEARCH

**Undergraduate Capstone Project - Aerial Image Processing** Aug 2022 - May 2023
*Computer Vision Researcher* *Worcester, MA*

- Developed an **end-to-end computer vision pipeline for GPS-denied aerial localization**, training a **CNN-based regression model** to estimate positional shifts from parachute imagery.
- Applied **SegFormer transformer-based semantic segmentation** and **OpenCV contour/edge detection** to perform real-time terrain classification and extract geographic boundaries from aerial video.

## PUBLICATIONS

Riaz, S., Zheng, H., **Roche, A.**, Zhang, M., Ragu, A., Penachio, S., Mukherjee, K., & Jonelagadda, A. (2026). *Multi-Task Learning for Non-Canonical Phoneme Recognition via Articulatory Feature Decomposition*. arXiv preprint arXiv:2608.22273.

Markus, A., **Roche, A.**, Ngan, C.-K., Cheung, Y.-T., & Prifti, K. (2022). *A Prognostic Machine Learning Framework and Algorithm for Predicting Long-term Behavioural Outcomes in Cancer Survivors.* DOI: 10.5220/0010893700003123

## SKILLS

**Languages:** Python, SQL, Java, R, Scala, C++
**ML Frameworks / Libraries:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV
**ML Techniques:** Transformers, Multi-Task Learning, RAG, Diffusion Models, State-Space Models, Cross-Attention, CNNs
**Speech / Audio ML:** TTS, WavLM, Phoneme Recognition, CTC, Conditional Flow Matching
**ML Systems:** Edge Inference, Quantization, GPU Training & Inference, CUDA, Model Optimization
**Data / Cloud:** PySpark, dbt, Snowflake, AWS, GCP, BigQuery